

The Persuasion–Prediction Gap: Outcome-Grounded Auditing of LLM Judges on Market-Resolved Financial and Crypto Reasoning

Abstract

LLM-as-a-judge now underpins automated evaluation and decision-support pipelines, yet judges are validated against human preferences or against other models, never against whether the reasoning they score was actually right. Both anchors can be swayed by fluent, confident rationales. We study a setting where the outcome itself supplies the label: reasoning about future, market-resolved events in finance and cryptocurrency. We introduce OG-JUDGE, a label-free benchmark that pairs reasoning traces about resolved prediction-market and crypto events with their realized outcomes, yielding outcome-discordant pairs consisting of one eventually-correct and one eventually-wrong rationale per event. Two diagnostics score judges: Outcome-Discrimination Accuracy (ODA) and the Persuasion–Prediction Gap (PPG), which decomposes preference into a persuasion channel and an outcome channel. An identification result shows why ODA alone cannot be read as judge quality. Across four judges spanning three providers on 350 contamination-controlled events, judges detect injected verifiable defects at 0.94–0.99 accuracy yet prefer the eventually-correct rationale at only 0.46–0.50, at or below chance and far short of an analytic Bayes ceiling of 0.706. The outcome share of preference is ≈ 0 for all judges, and for one judge (Gemini 2.5 Flash) the persuasion-adjusted outcome coefficient is significantly negative. PPG is positive across the reliably-estimable configurations and largest in the high-narrative crypto subdomain. Neither reasoning mode nor four outcome-agnostic mitigations restore outcome discrimination: judges reward rationales that sound right over rationales that are right. We release the metric code, data, and auditing protocol at <https://github.com/KurbanIntelligenceLab/llm-judge-finance-audit>.

Keywords: LLM-as-a-judge; evaluation of intelligent systems; prediction markets; trustworthy AI; reasoning assessment;

1 Introduction

LLM judges now stand in for human evaluators across summarization, dialogue, coding, safety, and increasingly multi-step *reasoning* [1, 2]. They are also becoming load-bearing components of knowledge-based and decision-support systems, where an LLM judge gates retrieval-augmented answers, ranks candidate solutions, or supplies the reward signal that trains or selects another model; the trustworthiness of the overall system then inherits the trustworthiness of its judge. When a judge scores a chain of reasoning, the implicit promise is that higher-scored reasoning is *better* reasoning. A growing body of work shows this promise is fragile: judges exhibit a verbosity- and confidence-driven preference, scoring fluent, assertive rationales above tentative ones [1, 3, 4], and recent controlled studies find that fluent reasoning, or injected rhetoric, can make a judge accept an incorrect answer [5, 6]. But better is itself validated almost entirely against human raters or against a stronger reference model [2, 7, 8]. Both anchors share a blind spot: a human rater shown a confident, fluent, numerically dense rationale

about an uncertain future event has no more access to the truth than the judge does. If judges prefer rationales that are persuasive rather than predictive, neither human agreement nor model agreement can detect it, because the validators are subject to the same persuasion.

A second line of work removes the human anchor by judging against *objective ground truth* [7, 9], but it does so on static tasks (math, coding, knowledge QA) whose answers exist at write time. There, the answer can leak into pretraining, persuasiveness is a post-hoc rationalization of a known result, and reasoning quality is confounded with answer recall. The question we ask needs ground truth that is both objective *and* unavailable when the rationale is written.

We ask a question that human-grounded and model-grounded protocols structurally cannot answer (Figure 1): *when a judge ranks reasoning about a question whose answer is revealed only after the fact, does the judge prefer the reasoning that turned out to be right, or the reasoning that merely sounds right?*

Why finance and crypto. Forecasting questions on regulated prediction markets (e.g., Kalshi, Polymarket) and cryptocurrency price-move events have three properties we exploit. Each event resolves to a verifiable 0/1, so correctness is automatic and needs no annotation. The answer also does not exist when the rationale is written, which means persuasiveness reflects an author’s epistemic style rather than a rationalization of a known result, and the outcome label cannot have leaked into pretraining [10, 11]. Finally, the domain offers a controllable range of narrative pressure: high-narrative crypto markets tend to attract more fluent, confident, plausible-but-wrong reasoning than base-rate-anchored macro or sports markets, giving a within-benchmark axis along which the gap may vary and which we test directly.

Outcome-grounded evaluation has stopped at the forecaster. A third, fast-growing line grounds LLM *forecasters* in realized market outcomes, scoring accuracy, calibration, and profitability [10–14]. Cheng et al. [10] benchmark LLM *forecasters* on live Polymarket data along directional accuracy, calibration, and simulated trading return; Nel [11] reports a reasoning paradox in which extended reasoning *worsens* calibration. Yet even these outcome-rich pipelines validate their *reasoning judges* against humans (for example, the blinded human study in [12]) or against synthesized reference traces [15], never against the very outcomes they already hold. The present work occupies the intersection these lines leave empty. We differ from prior work in four respects: we audit the *judge* of reasoning rather than the forecaster; we ground the audit in *realized future-market outcomes*, so the labels are contamination-free and carry genuine ex-ante uncertainty, unlike static known-answer ground truth [6, 7]; we use *naturally generated* discordant traces with no injected persuasion [6]; and we report a *persuasion-versus-outcome decomposition* with an accompanying identification result, rather than a single accuracy number or a manipulability demonstration.

Contributions.

1. **OG-JUDGE**, a label-free, contamination-resistant benchmark of reasoning traces about resolved market events, structured into *outcome-discordant pairs* and a parallel *verifiable-error* positive-control set (Section 3).
2. **Two diagnostics**: ODA (does the judge prefer the eventually-correct rationale?) and PPG (how much of the judge’s preference is persuasion vs. outcome?), with an identification result (Proposition 2) showing why both are needed.

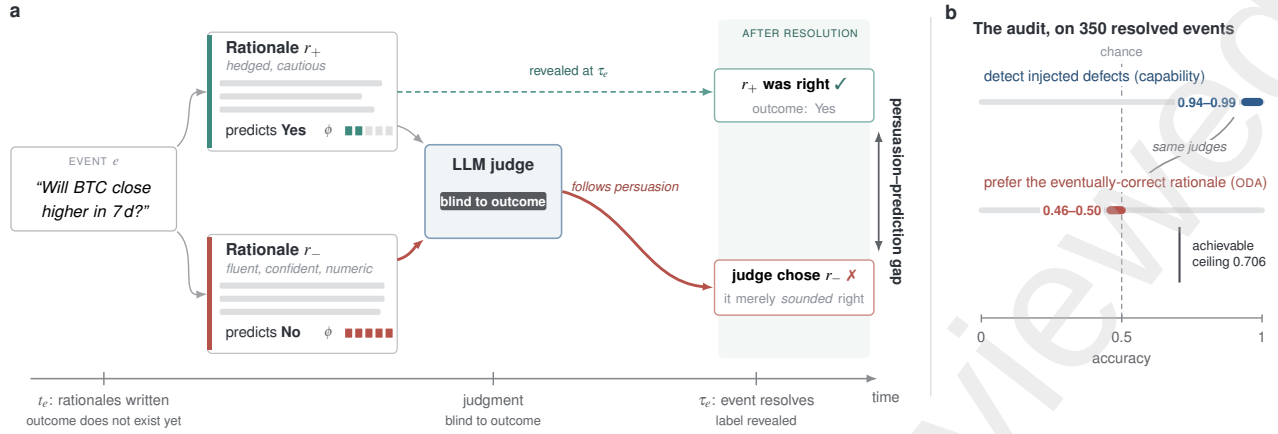


Figure 1: **The persuasion–prediction gap.** (a) The audit design follows the timeline of a market-resolved event. At write time t_e two rationales about the same future event disagree: the hedged r_+ (low surface persuasion ϕ) predicts Yes and the fluent, confident r_- (high ϕ) predicts No; the outcome does not yet exist, so neither writer nor judge can know it. An LLM judge, shown both but blind to the outcome, follows the persuasion signal and prefers r_- (red). When the event resolves at τ_e (teal, dashed), r_+ is vindicated. The gap is the distance between what sounded right and what was right; because the label comes from the realized outcome rather than from humans, this failure is invisible to standard human- or model-grounded judge validation. (b) The measured result across four judges spanning three providers on 350 contamination-controlled events (Table 3): the same judges that detect injected verifiable defects at 0.94–0.99 prefer the eventually-correct rationale at only 0.46–0.50, at or below chance (0.5, dashed) and far short of the model-based achievable ceiling of 0.706 (Section 5).

3. **A preregistered protocol** with order/length debiasing, a positive control that separates markets are hard from judges are weak, bootstrap CIs and permutation tests, and a crypto-vs-macro subdomain analysis (Section 5).
4. **Released metric code** and a documented benchmark-construction protocol (dataset assembly, temporal-cutoff enforcement, and the diagnostics); the method extends to other domains with verifiable outcomes.

To our knowledge, this is the first audit of LLM judges *of reasoning* that grounds the audit in *realized future-market outcomes on naturally generated outcome-discordant pairs* and reports a *persuasion-versus-outcome identification result*; a date-aware prior-art sweep (through mid-2026) found active work on judge reliability and bias but none combining these four elements. The closest concurrent work audits LLM judges of investment rationales before returns are observable but, by design, gates on inter-judge *agreement* and injected adversarial controls rather than grounding the audit in *realized outcomes* [16]; we detail this and a related downstream-outcome check [17] in Section 2.

2 Related Work and Positioning

LLM judges, their biases, and ground-truth judging. Judges approximate human preferences on open-ended tasks [1, 2] but exhibit systematic, content-independent biases: verbosity, position, self-enhancement, authority/confidence (tone) preference, and bandwagon effects [1, 3, 4, 18, 19]. Adversarial work shows

judges can be *tricked* into inflating scores for incorrect math solutions via injected rhetorical strategies [6], that the fluency of a reasoning chain biases judgments of answer factuality [5], and that a confident-but-flawed justification sways judges that skim for conclusions [20]. A parallel line removes the human anchor by scoring judges against *objective ground truth* [7, 9], arguing human preferences are unreliable indicators of correctness. These studies measure susceptibility against *known* answers (math/QA, write-time-known) or via *injected* persuasion. We differ on both axes: our rationales are *naturally generated* (no injection) and concern *future* events whose answers do not exist at write time, so persuasiveness is authentic and labels are contamination-free; and we *decompose* preference into persuasion vs. outcome rather than only demonstrating manipulability or reporting a single agreement number.

Market-grounded evaluation of forecasters. Prediction markets give contamination-proof labels and are now used to benchmark LLM *forecasters* on accuracy, calibration, and profitability [10–14, 21]. Nel [11] reports a reasoning paradox in which extended reasoning *worsens* calibration. All of this evaluates the entity that *makes* the prediction. We instead evaluate the entity that *judges* the reasoning, an orthogonal and under-studied target; we reuse the same label source but ask a different question.

Judging reasoning traces, and statistical identification. Ahamed et al. [15] use an LLM judge to score forecasting rationales against synthesized reference traces; Yang et al. [12] validate their reasoning judge against blinded human ratings; Molfese et al. [8] judge reasoning traces with human step-level annotations. None measures whether the judge’s preference tracks the *realized outcome* on outcome-discordant pairs, nor defines a persuasion–prediction decomposition. On the statistical side, Xu et al. [22] identify model rankings from judge comparisons *without* ground truth; we instead identify a judge’s outcome sensitivity *with* ground truth by separating it from persuasion. We also address a distinct contamination channel, preference leakage between related generator and evaluator models [23]: the two traces in every pair come from different writers, writer identity is absorbed as a fixed effect, and, because two of our four judges share a model family with the writer roster, we verify that the conclusions are unchanged on the judge-specific subsets of pairs containing no trace written by the judge’s own model (see Limitations). Auditing the judge against the outcomes it is blind to is the open problem we address.

Concurrent auditing of judges against outcomes. Two concurrent efforts move judge validation toward downstream ground truth and are our closest neighbors. Chang et al. [16] audit LLM judges of investment rationales in the pre-realization window, but by design decline to ground on realized returns (treating them as too delayed and noisy) and instead gate on inter-judge weighted agreement plus injected adversarial controls in a controlled capital-allocation prototype; they report that terse-correct rationales are penalized relative to more verbose ones, corroborating our persuasion channel from a different method. Chen et al. [17] test the criterion validity of an LLM-judge rubric against verified business conversion, again a downstream-outcome check but outside forecasting and without a persuasion/outcome decomposition. OG-JUDGE differs from both: it grounds the audit in *realized market outcomes* rather than inter-judge agreement or a static rubric; it uses *naturally generated outcome-discordant pairs* rather than injected controls; and it supplies a *persuasion-versus-outcome identification result* that neither provides. Table 1 places OG-JUDGE against the closest neighbors along the dimensions that define the contribution.

Table 1: **Closest prior work and what each leaves open.** ● yes ◐ partial ○ no. The union of open circles down a column is the contribution OG-JUDGE fills.

Work	<i>what is evaluated</i>		<i>benchmark design & analysis</i>		
	Judges reasoning (not f’cast)	Realized outcome label	Outcome- discordant pairs	Natural traces (no inj.)	Persuasion vs. outcome decomp. (PPG)
Trick the Grader [6]	●	○	○	○	○
Reasoning chains [5]	●	○	○	●	○
JudgeBench [7]	●	◐ [†]	○	○	○
TFRBench [15]	●	●	○	●	○
Prophet Arena [12]	◐	●	○	●	○
KalshiBench [11]	○	●	○	●	○
PolyBench [10]	○	●	○	●	○
Judge-bias [3]	●	○	○	○	○
ReTraceQA [8]	●	○	○	●	○
OG-JUDGE (ours)	●	●	●	●	●

[†]JudgeBench uses *static, write-time-known* ground truth (math, coding, knowledge), which is contaminable and lacks ex-ante uncertainty; OG-JUDGE uses *realized future-market* outcomes.

3 The OG-JUDGE Benchmark

Events. We collect resolved binary events from three sources: (a) regulated/decentralized prediction markets (Kalshi, Polymarket) via public APIs; (b) crypto price-move events constructed automatically from public OHLCV data: “will asset a ’s price close at least $\theta\%$ relative to its level at timestamp t , h hours later?”, with the per-event threshold θ fixed at event creation from the symmetric grid $\{-8, -4, -3, -1.5, 0, +1.5, +3, +4, +8\}$ and horizons $h \in \{24, 168\}$ (the θ referenced by the edge proxy of App. A.6b is this stated threshold); (c) macro/economic events templated from public time series (e.g., FRED), following Karger et al. [13]. Each event e has a realization $y_e \in \{0, 1\}$ known only at resolution time $\tau_e > t_e$.

Traces. For each event we elicit rationales from a panel of *writer* models $\mathcal{W}$ (three model sizes of a single proprietary family, each in two reasoning variants) using a fixed template that returns a free-text rationale and an explicit probability $\hat{p}(r) \in [0, 1]$ and label $\hat{y}(r) = \mathbf{1}[\hat{p}(r) \geq 0.5]$. Writers see only information available at t_e (we enforce temporal cutoffs and, for crypto, truncate price history at t_e). Correctness is $c(r) = \mathbf{1}[\hat{y}(r) = y_e]$.

141 **Outcome-discordant pairs.** For an event e we form pairs (r_+, r_-) with $c(r_+) = 1, c(r_-) = 0$, drawn from
 142 *different* writers to avoid stylistic collapse, matched on event so the question is held fixed. The judge sees both
 143 (order randomized) and chooses the better-reasoned trace, *blind* to y_e .

144 **Verifiable-error positive control.** In parallel we build pairs where r_- contains a *demonstrable* defect injected
 145 into an otherwise-matched correct rationale: an arithmetic error, a self-contradiction, or a claim that contradicts
 146 a value present in the provided context. These pairs have ground truth about *reasoning quality* independent of
 147 market noise. A judge that fails here cannot detect even blatant defects; a judge that succeeds here but is at
 148 chance on discordant pairs has a *predictive-validity* blind spot specifically. This control is what lets us separate
 149 markets are hard from judges are weak.

150 **Persuasion features.** For each trace we compute a vector $\phi(r) \in \mathbb{R}^k$ of *automatic, content-independent*
 151 surface features ($k=8$; Table 2); none requires another LLM, avoiding circularity. Pairing already differences
 152 out event-level location; in the fitted model the per-feature gaps $\Delta\phi$ are additionally standardized to zero mean
 153 and unit variance across pairs, as in the released code.

154 4 Diagnostics and Identification

155 **Definition 1** (Outcome-Discrimination Accuracy). For judge J and a set $\mathcal{P}$ of outcome-discordant pairs,
 156 $\text{ODA}(J) = \frac{1}{|\mathcal{P}|} \sum_{(r_+, r_-) \in \mathcal{P}} \mathbf{1}[J \text{ prefers } r_+]$. Chance is 0.5. An AUC variant uses pointwise judge scores $s_J(r)$ vs.
 157 correctness labels.

158 **Setup and assumptions.** Fix an outcome-discordant pair (r_+, r_-) with r_+ eventually correct. Let presentation
 159 order be randomized and let $z \in \{0, 1\}$ indicate whether the correct trace is shown first. Write $d_+ = \phi(r_+) -$
 160 $\phi(r_-) \in \mathbb{R}^k$ for the *correct-minus-wrong* persuasion gap, and let $\Delta\phi = \phi(r_{\text{first}}) - \phi(r_{\text{second}}) = (2z - 1)d_+$ be the
 161 displayed gap. We model the judge’s choice by the paired logistic form in Eq. (1) below, with linear persuasion
 162 index $u = \beta_\phi^\top d_+$. We use three assumptions, stated where needed: **(A1)** order randomization, $z \perp d_+$ with
 163 $\Pr[z=1] = \frac{1}{2}$; **(A2)** *overlap/non-degeneracy*: the design $(1, z, \Delta\phi)$ has full column rank; under (A1) this holds
 164 if and only if no nonzero linear combination of the persuasion gaps is almost surely constant across pairs,
 165 i.e. $\text{Var}(\gamma^\top d_+) > 0$ for every $\gamma \neq 0$ (equivalently, $\text{Cov}(d_+) \succ 0$); it fails in particular when persuasion is a
 166 deterministic function of correctness, which forces d_+ to be constant across pairs; **(A3)** the choice probability
 167 follows Eq. (1), or we read its coefficients as the best logistic projection. (A1) is enforced by design; (A2) is
 168 the identifying condition we return to; (A3) is the working model.

Table 2: **Automatic persuasion features** $\phi(r)$, grouped by channel. All are lexical or surface statistics computed without any LLM, which keeps the persuasion measure transparent and non-circular.

Feature	Notation	Operationalization
VERBOSITY		
Length	ϕ_{len}	token count of the rationale
Readability	ϕ_{read}	mean sentence length / syllable proxy
CONFIDENCE / TONE		
Booster rate	ϕ_{boost}	rate of confidence terms (<i>clearly, certainly</i>)
Hedge rate	ϕ_{hedge}	rate of hedges (<i>might, possibly, unclear</i>)
Sentiment	ϕ_{sent}	lexicon polarity of the rationale
FORM / SPECIFICITY		
Numeric density	ϕ_{num}	share of numeric / percent tokens
Entity / ticker density	ϕ_{ent}	share of capitalized / ticker-like tokens
Structure	ϕ_{struct}	count of list / step / discourse markers

169

170 **Definition 2** (Persuasion–Prediction Gap). *Fit the paired logistic model for the judge preferring the first trace,*

$$\Pr[J \text{ picks first} \mid z, \Delta\phi] = \sigma(\beta_0 + \beta_{\text{out}} z + \beta_{\phi}^{\top} \Delta\phi), \quad (1)$$

171 *with features standardized. Let S_{out} and S_{ϕ} be the McFadden pseudo- R^2 of the outcome-only model $\{1, z\}$*
 172 *and of the persuasion-only model $\{1, \Delta\phi\}$, each measured against the intercept-only null (App. A, A.3; their*
 173 *difference is attribution-order-independent, App. A.7). Define $\text{PPG}(J) = S_{\phi} - S_{\text{out}}$. Here β_{out} is outcome*
 174 *sensitivity (preference for the correct trace after controlling for persuasion) and β_{ϕ} is persuasion sensitivity.*

175 **The diagnostic problem, made exact.** Our first result shows precisely what ODA measures and why it
 176 conflates the two channels. Let $\Delta_{\text{out}} = \frac{1}{2} \mathbb{E}_u[\sigma(\beta_{\text{out}} + u) - \sigma(u)]$ and $\Delta_{\text{pers}} = \mathbb{E}_u[\sigma(u) - \frac{1}{2}]$.

177 **Proposition 1** (Exact ODA decomposition). *Under (A1) and (A3), with $\beta_0=0$ to isolate the outcome and*
 178 *persuasion channels (the general position-bias identity, in which β_0 is separately identified and conditioned*
 179 *out, is in App. A),*

$$\text{ODA}(J) - \frac{1}{2} = \underbrace{\Delta_{\text{out}}}_{\text{outcome sensitivity}} + \underbrace{\Delta_{\text{pers}}}_{\text{persuasion–correctness coupling}}. \quad (2)$$

180 *Consequently: (i) if $\beta_{\text{out}}=0$ then $\text{ODA} - \frac{1}{2} = \Delta_{\text{pers}}$, so any departure of ODA from chance is pure persuasion*
 181 *artifact; (ii) if the persuasion gap is symmetric about zero, $d_+ \stackrel{d}{=} -d_+$ (as when persuasion is independent*
 182 *of correctness and the two pair members are exchangeable), then $\Delta_{\text{pers}}=0$ and $\text{ODA} - \frac{1}{2} = \Delta_{\text{out}}$ reflects only*
 183 *outcome sensitivity; (iii) a judge with genuine sensitivity ($\beta_{\text{out}} > 0$) can still have $\text{ODA} = \frac{1}{2}$ when $\Delta_{\text{out}} = -\Delta_{\text{pers}}$.*
 184 *Hence ODA alone does not identify judge quality; PPG and β_{out} do.*

185 *Proof.* Condition on the pair, hence on u . Under (A3) with $\beta_0=0$: if $z=1$ the correct trace is first and is preferred
 186 with probability $\sigma(\beta_{\text{out}} + u)$; if $z=0$ it is second, so it is preferred with probability $1 - \sigma(-u) = \sigma(u)$. By
 187 (A1), $\mathbb{E}[\mathbf{1}\{J \text{ prefers } r_+\} \mid u] = \frac{1}{2} \sigma(\beta_{\text{out}} + u) + \frac{1}{2} \sigma(u)$. Taking expectations over u and subtracting $\frac{1}{2}$ splits into

188 $\frac{1}{2} \mathbb{E}[\sigma(\beta_{\text{out}} + u) - \sigma(u)] + \mathbb{E}[\sigma(u) - \frac{1}{2}]$, which are Δ_{out} and Δ_{pers} . For (i), set $\beta_{\text{out}}=0$. For (ii), $d_+ \stackrel{d}{=} -d_+$ gives
189 $u \stackrel{d}{=} -u$, and $\sigma(u) + \sigma(-u) = 1$ yields $\mathbb{E}[\sigma(u)] = \frac{1}{2}$. (iii) is immediate from (2). The general $\beta_0 \neq 0$ identity is
190 in App. A. $\square$

191 **Proposition 2** (Identification of outcome sensitivity). *Under (A1) and (A3), the coefficients $(\beta_0, \beta_{\text{out}}, \beta_\phi)$ are*
192 *identified if and only if (A2) holds. When identified, β_{out} is the persuasion-adjusted log-odds of preferring*
193 *the correct trace, and $\beta_{\text{out}} > 0$ iff, holding $\Delta\phi$ fixed and averaging over the randomized presentation order,*
194 *the judge prefers the correct trace above chance. If (A2) fails, i.e. some nonzero combination $\gamma^\top d_+$ is a.s.*
195 *constant, canonically when $d_+ \equiv \kappa$ a.s. for a fixed vector κ (as when persuasion is a deterministic function of*
196 *correctness, whence $\Delta\phi = (2z - 1)\kappa$ is affine in z), the outcome and persuasion channels are confounded and*
197 *$\beta_{\text{out}}, \beta_\phi$ are not separately identified.*

198 *Proof.* GLM coefficients are identified iff the design has full column rank, which is (A2). Under (A1) a
199 nontrivial dependence $a + bz + \gamma^\top \Delta\phi = 0$ a.s. forces, on the events $\{z=1\}$ and $\{z=0\}$ (each of probability $\frac{1}{2}$,
200 with d_+ distributed identically on both by $z \perp d_+$), $\gamma^\top d_+ = -(a + b)$ and $\gamma^\top d_+ = a$ a.s.; since $a + bz = 0$ a.s.
201 alone forces $a = b = 0$, (A2) fails exactly when some $\gamma^\top d_+$ with $\gamma \neq 0$ is a.s. constant. The canonical failure
202 is $d_+ \equiv \kappa$ a.s.: then $\Delta\phi = (2z - 1)\kappa = 2\kappa z - \kappa \in \text{span}\{1, z\}$, so the persuasion columns are redundant and
203 the outcome and persuasion channels are perfectly confounded. Conversely, when (A2) holds the coefficients
204 are identified; given (A1), β_{out} is the coefficient on z holding $\Delta\phi$ fixed, i.e. the persuasion-adjusted outcome
205 association, and, since under (A1) the order-averaged probability of preferring the correct trace at fixed
206 $\Delta\phi = \delta$ is $\frac{1}{2} [\sigma(\beta_0 + \beta_{\text{out}} + \beta_\phi^\top \delta) + 1 - \sigma(\beta_0 + \beta_\phi^\top \delta)]$, monotonicity of σ gives the sign claim. Regularity for
207 the estimator is in App. A. $\square$

208 Proposition 2 is also a design rule: the benchmark must contain pairs in which the correct trace is sometimes
209 more and sometimes less persuasive than the wrong one, or the two channels cannot be separated. We verify this
210 holds in our construction and confirm on synthetic data that the estimator recovers a planted persuasion-driven
211 judge ($\text{ODA} \approx \frac{1}{2}$, $\text{PPG} > 0$, $\beta_{\text{out}} \approx 0$) and a planted outcome-driven judge ($\text{ODA} \gg \frac{1}{2}$, $\text{PPG} < 0$, $\beta_{\text{out}} > 0$).

212 **Inference.** We report β_{out} , $\|\beta_\phi\|$, S_ϕ , S_{out} , PPG, a likelihood-ratio test for $\beta_{\text{out}}=0$, a permutation test that
213 shuffles z , and percentile bootstrap CIs resampling *events* (not pairs) to respect clustering. Because S_ϕ
214 aggregates a multi-dimensional block and is mechanically inflated by its dimension, we calibrate every share
215 against a permutation null (App. A); the test on β_{out} is the inferential anchor for outcome sensitivity.

216 **Separating hard markets from weak judges.** Three controls. (i) The *verifiable-error* set upper-bounds the
217 judge's ability to detect clear defects; chance there indicts the judge directly. (ii) *Edge stratification*: restrict
218 to events where the correct call was identifiable ex-ante, using an edge proxy computed only from t_e -time
219 (pre-resolution) information: the t_e -time price-implied probability where available, verified at the code level to
220 use only pre- t_e information (no use of the closing price or any post-resolution signal, which would leak the
221 outcome); persistent chance ODA even at high ex-ante edge refutes the pure luck reading. (iii) The *primary*
222 *claim* is about PPG (persuasion explains much, outcome little), which is informative regardless of whether
223 markets are predictable: even granting that the outcome signal is small, a good judge should not substitute
224 persuasion for it.

225 **Confound control.** Order is randomized per pair; we additionally report a length-matched subset to neutralize
226 verbosity [3], and we include the writer-model identity as a fixed effect to absorb stylistic confounds. A position-

swapped replicate (re-running the three position-bias-degenerate configurations with trace order reversed) would let us distinguish genuine positional preference from a harness artifact directly rather than via the indirect verifiable-error comparison in App. A.8, but was not run in this study; we flag it as a natural follow-up rather than claim it here.

5 Experimental Protocol and Results

We preregistered hypotheses H1–H4 and the analyses that test them below. All results are reported on the pooled event set of 350 contamination-controlled resolved events, with per-subdomain breakdowns where relevant.

Models. *Judges* $\mathcal{J}$: four proprietary API judges spanning three providers: Claude Haiku 4.5 and Claude Sonnet 4.5 (Anthropic), GPT-4o-mini (OpenAI), and Gemini 2.5 Flash (Google), accessed via the OpenRouter API on 2026-07-01 (run-manifest date) under the model identifiers `anthropic/claude-haiku-4.5`, `anthropic/claude-sonnet-4.5`, `openai/gpt-4o-mini`, and `google/gemini-2.5-flash`, at judge temperature 0.0 (deterministic decoding). Each judge is evaluated in a non-reasoning and a reasoning (thinking) variant. *Writers* $\mathcal{W}$: three model sizes spanning a size range, Claude Haiku 4.5 (small), Claude Sonnet 4.5 (mid), and Claude Opus 4.1 (large), each writing in both a non-reasoning and a reasoning variant at sampling temperature 0.7 (max tokens 280 non-reasoning, 480 reasoning). Two of the four judges (Haiku 4.5 and Sonnet 4.5) coincide with writer models; the self-preference implications and a robustness slice excluding own-model traces are reported in Limitations. The original protocol (runbook) called for 5 repeated seeds per stochastic judge call; this was not executed as planned: judges were run once each at temperature 0.0 (a deterministic setting, for which repeated seeds would add no variance beyond API-level nondeterminism), and all reported uncertainty is event-clustered bootstrap over the pair sample, not seed-repetition variance. We flag this as a deviation from the original protocol rather than silently reporting seed-based uncertainty that was not in fact collected.

Hypotheses. H1 (gap exists): $\text{PPG}(J) > 0$ with $S_\phi \gg S_{\text{out}}$ for most judges. H2 (chance discrimination): $\text{ODA}(J) \approx 0.5$ on discordant pairs while the verifiable-error control is $\gg 0.5$. H3 (reasoning paradox): reasoning judges do not reduce, and may increase, PPG relative to their non-reasoning variants [11]. H4 (narrative widens the gap): PPG and the persuasion sensitivity are larger in crypto/high-narrative subdomains than in macro/sports.

Positive control (harness validity). We first verified the harness on the verifiable-error set ($\sim 1\text{k}$ injected-defect pairs per judge; Table 3). With a defect-naming prompt, all four judges detected the injected defect at 0.94–0.99 accuracy (Table 3, VE column), so the judges are individually capable and the pairwise harness is sound. This is the denominator that turns a chance-level ODA into evidence about outcome-blindness specifically, rather than about judging ability in general.

Outcome discrimination (H2). On the outcome-discordant pairs, with presentation order randomized, pooled ODA was 0.46–0.50 across the four judges (Table 3), at or nominally below chance (the event-clustered bootstrap CIs straddle or sit just below 0.5), while the same judges scored 0.94–0.99 on the verifiable-error control. Judges that reliably catch blatant defects cannot separate the eventually-correct forecast from the eventually-wrong one (Figure 2).

For three of the four non-reasoning judges, the likelihood-ratio/ z -permutation test for β_{out} does not reject zero outcome sensitivity: Haiku 4.5 (ODA= 0.501, CI[0.452, 0.546], binomial $p = 0.97$ vs. chance; LR= 0.84, permutation $p = 0.35$), Sonnet 4.5 (ODA= 0.461, CI[0.424, 0.500], binomial $p = 0.04$; LR= 0.12, permutation $p = 0.76$), and GPT-4o-mini (ODA= 0.460, CI[0.417, 0.507], binomial $p = 0.03$; LR= 3.39, permutation $p = 0.08$). Gemini 2.5 Flash is the exception: its permutation test on β_{out} rejects the null (LR= 8.06, permutation $p=0.0033$ uncorrected; it survives a four-way Holm correction, whose step-down threshold for the smallest p is $0.05/4 = 0.0125$), and its estimated $\hat{\beta}_{\text{out}} = -0.518$ is *negative*: after controlling for persuasion, Gemini's residual outcome sensitivity runs in the wrong direction, systematically favoring the eventually-*incorrect* trace slightly more than chance would predict (ODA= 0.468, CI[0.422, 0.508]), rather than showing any positive discrimination. The other three judges' point estimates for $\hat{\beta}_{\text{out}}$ are also negative (Haiku -0.174 , Sonnet -0.115 , GPT-4o-mini -0.335), though not distinguishable from zero at conventional significance; taken together, the sign pattern across all four judges is consistent with weak anti-outcome coupling rather than any judge showing positive outcome discrimination. The binomial test on ODA itself (Sonnet $p = 0.04$, GPT-4o-mini $p = 0.03$) shows a nominal deviation below chance that the bootstrap CI, which straddles 0.5 for both, does not by itself flag; we treat the permutation test on β_{out} , which directly targets the coefficient rather than ODA, as the more decisive test: it does not reject zero for these two judges, so we read their below-chance binomial deviations as nominal and reserve the claim of a statistically significant (negative) outcome coupling for Gemini 2.5 Flash alone. This supports H2.

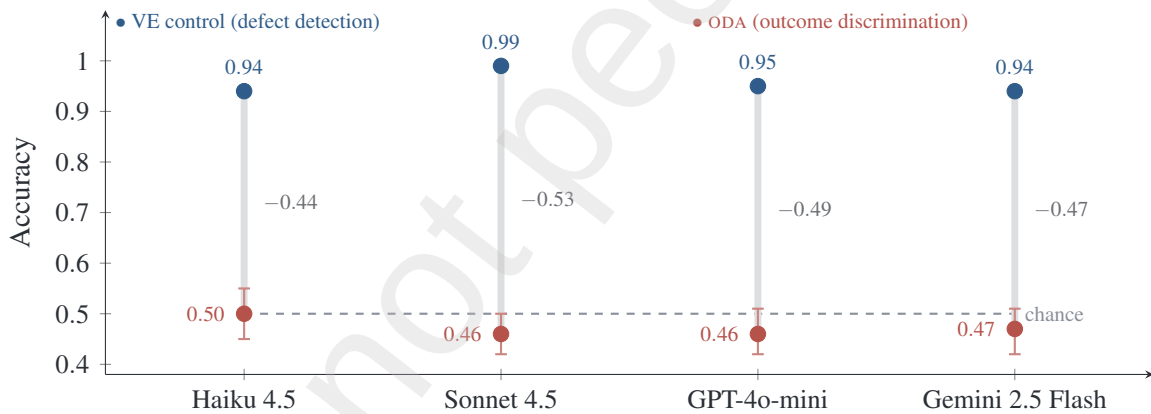


Figure 2: **The dissociation (H2).** Each judge is a vertical pair: a blue point for verifiable-error accuracy (defect detection) and a red point for outcome-discrimination accuracy (ODA), joined by a grey stem whose length (the labelled drop) is the collapse from near-ceiling capability to chance. Defect detection sits at 0.94–0.99; ODA sits at 0.46–0.50, on the chance line (0.5, dashed). Red whiskers are 95% event-clustered bootstrap CIs on ODA. The same judges that catch injected defects almost perfectly cannot tell the eventually-correct forecast from the eventually-wrong one. ODA point estimates and CIs match the non-reasoning rows of Table 3 exactly (event-clustered bootstrap, $n_{\text{boot}}=800$).

Persuasion–prediction decomposition (H1). Fitting Eq. (1), the outcome share S_{out} was ≈ 0 for every judge while the persuasion share S_{ϕ} ranged 0.03–0.30, giving $\text{PPG} > 0$ for the three non-degenerate base judges (Haiku 4.5, GPT-4o-mini, Gemini 2.5 Flash: 0.22–0.30; Table 3; Figure 3). Sonnet 4.5 base's nominal $\text{PPG} = 0.03$ carries the position-bias caveat noted in Table 3's footnote (its 97.7% first-pick rate makes $S_{\text{out}}, S_{\phi}, \text{PPG}$ unreliable for this cell; App. A.8), so we do not read it as an independent, small-but-genuine gap. The likelihood-ratio and z -permutation tests for β_{out} do not reject zero outcome sensitivity for three of the

four base judges (Haiku 4.5, Sonnet 4.5, GPT-4o-mini; permutation $p \in [0.08, 0.76]$); Gemini 2.5 Flash is the exception (permutation $p=0.003$), consistent with a small but nonzero S_{out} . Preference is otherwise explained by the automatic persuasion features, not by which trace was correct, supporting H1, a conclusion that stands comfortably on the three non-degenerate judges alone and does not require Sonnet's contaminated cell. The gap is largest for Haiku 4.5 (0.30) among the reliably-estimable configurations.

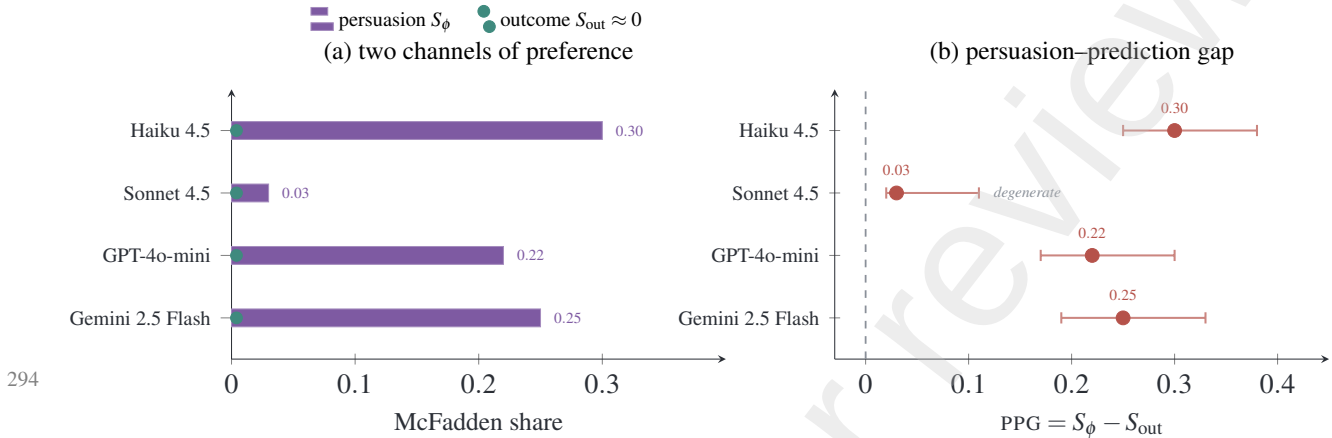


Figure 3: Persuasion vs. outcome shares (H1). (a) For each non-reasoning judge the persuasion share S_ϕ (purple bar) is sizeable, while the outcome share S_{out} (teal dot; drawn at a nominal 0.004 offset for visibility; the fitted values are 0.000–0.003, Table 3) is pinned at ≈ 0 . (b) The resulting Persuasion–Prediction Gap $\text{PPG} = S_\phi - S_{\text{out}}$ is positive for the three non-degenerate judges, 0.22–0.30 (points with 95% event-clustered bootstrap CIs; dashed line marks zero). Sonnet 4.5’s nominal $\text{PPG} = 0.03$ is shown for completeness but is position-bias-degenerate (Table 3, App. A.8) and excluded from the H1 reading. PPG point estimates and CIs match the non-reasoning rows of Table 3 exactly (event-clustered bootstrap, $n_{\text{boot}}=800$).

Reasoning judges (H3). Enabling each judge’s reasoning (thinking) mode did not restore outcome discrimination: ODA stayed at 0.45–0.46 across all four judges (Table 3; pooled event-clustered bootstrap, $n \approx 744$ per judge), never approaching the verifiable-error ceiling. S_{out} stayed small for three of four judges (Haiku 4.5 0.005, GPT-4o-mini 0.003, Gemini 2.5 Flash 0.003); Sonnet 4.5’s reasoning variant is the exception ($S_{\text{out}} = 0.040$, still small in absolute terms but an order of magnitude above the other three, though this cell is itself position-bias-degenerate, see below). Three of the eight judge $\times$ variant cells that feed this comparison (Judge A reasoning, 100% first-pick; Judge B base, 97.7%; Judge B reasoning, 99.5%) are the position-bias-degenerate configurations flagged in Table 3 (App. A, A.8), so their PPG values reflect near-constant positional picking rather than a genuine persuasion-vs-outcome trade-off; we exclude them from the across-configuration H3 comparison below. Restricting to the two judges with clean cells on *both* variants (GPT-4o-mini and Gemini 2.5 Flash), the preregistered form of H3 (that reasoning would *not* reduce, and might increase, PPG) was *not* supported: reasoning left PPG essentially unchanged for GPT-4o-mini ($0.22 \rightarrow 0.22$) and *lowered* it for Gemini 2.5 Flash ($0.25 \rightarrow 0.16$). The Gemini reduction came from weaker persuasion sensitivity (a smaller S_ϕ), not from recovered outcome sensitivity: S_{out} and ODA did not move toward the ceiling in either judge. Reasoning thus attenuates reliance on surface persuasion for Gemini 2.5 Flash, leaves it unchanged for GPT-4o-mini, and supplies none of the missing outcome signal in either case, so the core failure is unchanged. The Haiku ($0.30 \rightarrow 0.06$) and Sonnet ($0.03 \rightarrow -0.02$) figures in Table 3 are directionally consistent with this same pattern but should not be read as independent confirmations, since one or both cells in each pair are

position-bias-degenerate. We report H3 as refuted in its preregistered (PPG) form on the evidence that is not confounded by position bias, while its motivating observation, that reasoning does not fix outcome-blindness, is directly supported for GPT-4o-mini and Gemini 2.5 Flash and not contradicted by the remaining two judges (whose ODA also stays at or below chance, even though the underlying PPG decomposition is unreliable for those cells).

Subdomain and edge stratification (H4). PPG was numerically largest in the high-narrative crypto subdomain (0.20, 95% CI [0.17, 0.24]) and lower across prediction markets (0.15), macro (0.14) and sports (0.15), but the 95% event-clustered bootstrap CIs of the four subdomains overlap (Figure 4). We therefore report H4 as directionally supported but not statistically established at this sample size: the gap is largest where narrative pressure is highest, yet the lower-narrative subdomains do not resolve into a finer gradient. In the crypto subdomain, where an ex-ante edge proxy is available, we tercile-stratify the 388 discordant pairs per judge by edge magnitude ($n=180$ low, 92 mid, 116 high) and pool ODA on the high-edge tercile across all four judges ($n=464$ judge $\times$ pair observations): ODA = 0.44 (95% CI [0.395, 0.485]), significantly *below* chance (binomial $p = 0.01$ vs. 0.5; the CI excludes 0.5). Restricting to the events where the outcome was most ex-ante decidable does not raise ODA; the failure is judge outcome-blindness, not market unpredictability.

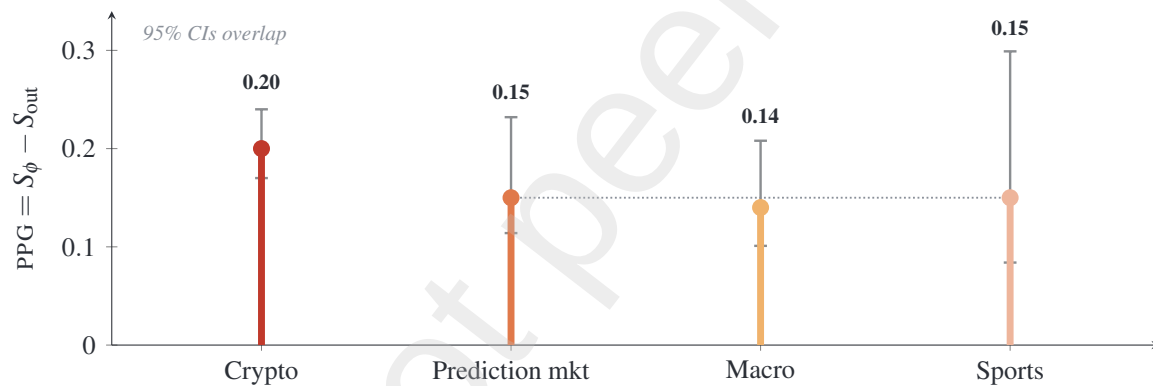


Figure 4: **Persuasion-Prediction Gap by subdomain (H4).** PPG pooled over the four judges. The gap is numerically largest in the high-narrative crypto subdomain (0.20) and lower across prediction markets (0.15), macro/FRED (0.14) and sports (0.15), but the 95% event-clustered bootstrap CIs of the four subdomains overlap. The data support the directional H4 claim (the gap is largest where narrative pressure is highest) but do not establish a monotone gradient among the lower-narrative domains at this sample size. Subdomains are ordered by narrative intensity (crypto highest); bar counts are judge $\times$ pair observations pooled over the four judges: crypto $n=1552$, prediction markets $n=352$, macro/FRED $n=809$, sports $n=256$. PPG 95% event-clustered bootstrap CIs: crypto [0.172, 0.240], prediction markets [0.114, 0.232], macro/FRED [0.101, 0.208], sports [0.084, 0.299], confirming the pairwise overlap noted above. A post-hoc, approximate crypto-vs-pooled-rest contrast is nominally significant but not preregistered (App. A, A.6); we retain the conservative directional reading here.

Outcome-agnostic mitigations. We tested four mitigations a practitioner might deploy on the crypto subdomain ($n=388$ discordant pairs); coverage is not uniform across judges, which we report explicitly rather than implying a full 4×4 sweep. The anti-persuasion rubric and self-critique were run for two judges (Haiku 4.5 and GPT-4o-mini): the rubric left ODA at 0.47–0.49 (Haiku 0.49, GPT-4o-mini 0.47), and self-critique left it

at 0.41–0.47 (Haiku 0.41, GPT-4o-mini 0.47); neither restored outcome discrimination. Self-critique in fact enlarged crypto-only PPG sharply for Haiku 4.5 (baseline PPG= 0.41 on this subdomain, distinct from its pooled-scale Table 3 value of 0.30, $\rightarrow$ 0.87 under self-critique); the persuasion share driving this increase is itself close to saturating ($S_\phi = 0.9985$), likely reflecting a near-degenerate fit under self-critique rather than a proportionally large behavioral shift, so we read this as a directional warning (self-critique can backfire) rather than a precise magnitude. Length/structure normalization was tested only for Haiku 4.5: restricting to pairs with a length gap of at most 50 words between the two rationales ($n=136$ of the full 388) gives ODA= 0.50, indistinguishable from the full-sample ODA= 0.4845 ($n=388$); length normalization does not recover outcome discrimination either. Four-judge majority-vote ensembling (base variants; $n=388$, crypto-only) gave ODA= 0.46, still at chance (crypto-only; the pooled ensemble in Table 3 gives 0.48, $n=737$). None of these mitigations restored outcome discrimination on the subdomain and judges tested. This is the expected signature of a failure that requires *outcome grounding* rather than better prompting: without access to the outcome, an evaluator cannot recover a signal it never observed.

Post-hoc exploratory validation: is the gap judge-blindness or market unpredictability? The tercile check above is coarse. To give the markets are simply hard objection its strongest, quantitative form, we compute an analytic Bayes-ODA ceiling: under a driftless-normal model of the t_e -time price series, the best-possible (oracle) discriminator attains ODA= 0.706 (95% CI [0.689, 0.724]) pooled over the crypto and macro domains (270 of the 350 events, for which a t_e -time price is available; the prediction-market domain has no recorded t_e -time price and is a data gap, not a null finding, so no ceiling is reported for it), far above the 0.45–0.50 observed for every judge (Table 3). Because this ceiling is itself model-conditional, we validate it empirically on the crypto subset ($n=192$), where we verified at the code level that our edge-proxy reconstruction uses only pre- t_e information (no leakage): the realized direction accuracy is 0.734 (binomial $p < 10^{-10}$ vs. chance), matching the analytic crypto ceiling of 0.718 to within sampling noise, and the underlying probabilities are well-calibrated (Brier score 0.173 vs. 0.25 for an uninformative constant-0.5 baseline; expected calibration error 0.064; Figure 5). We do not extend this empirical validation to the macro domain because our reconstruction of its edge formula matches the stored values only approximately (82% within a loose tolerance), so the macro and pooled ceilings remain model-conditional pending a verified macro edge formula, an open item we flag rather than claim as validated. Quantile-stratified ODA on the crypto subdomain (Figure 6) ranges [0.33, 0.59] across all eight judge $\times$ variant configurations across edge quartiles, and in the highest-edge quartile (Q4) stays in [0.39, 0.53] for every configuration: the 0.706 ceiling is never approached even where the ex-ante signal is strongest. Together these checks support the judge-blindness reading over a pure-luck alternative: the shortfall on the events where a ceiling can be computed reflects judges not using available signal, not events being intrinsically undecidable. Full data accounting, an order-independence (Shapley) check confirming PPG is invariant to attribution order, and the position-bias diagnosis referenced in Table 3’s footnotes are in Appendix A–A.8.

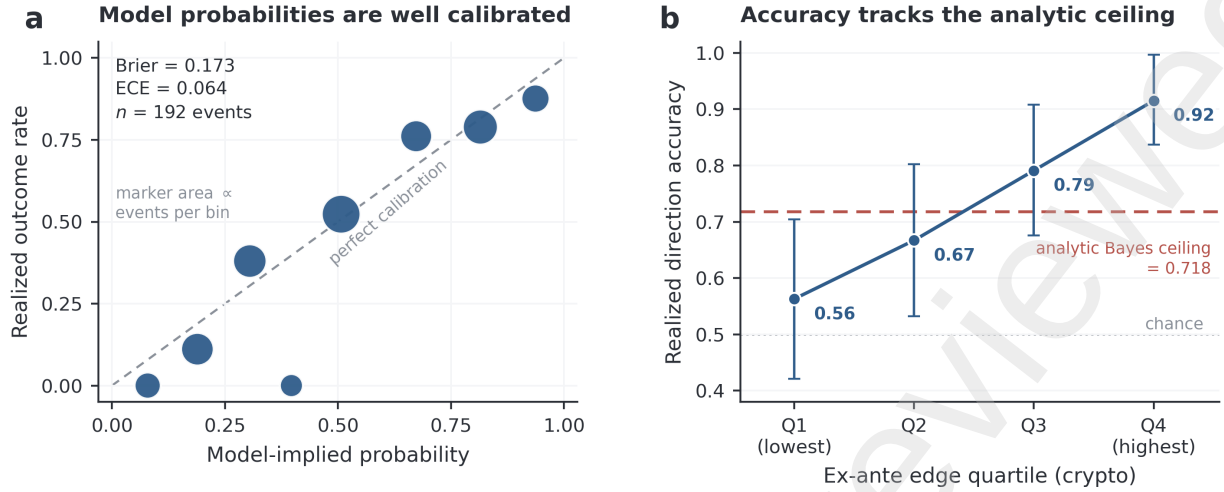


Figure 5: **Empirical validation of the analytic Bayes ceiling, crypto subset only.** Restricted to the 192 crypto events, where we verified at the code level that our edge-proxy reconstruction closely matches the released event data using only pre- t_e information (mean absolute deviation 0.006; App. A.6b). **(a)** The model-implied probabilities are well-calibrated: Brier 0.173 vs. 0.25 for an uninformative constant-0.5 baseline, ECE 0.064; marker area is proportional to the number of events per probability bin. **(b)** Realized direction accuracy rises monotonically with ex-ante edge, from 0.56 in the lowest edge quartile to 0.92 in the highest (whiskers, 95% CIs), tracking the analytic crypto ceiling of 0.718 (dashed), which confirms the ceiling is not a modeling artifact on the domain where it could be directly checked.

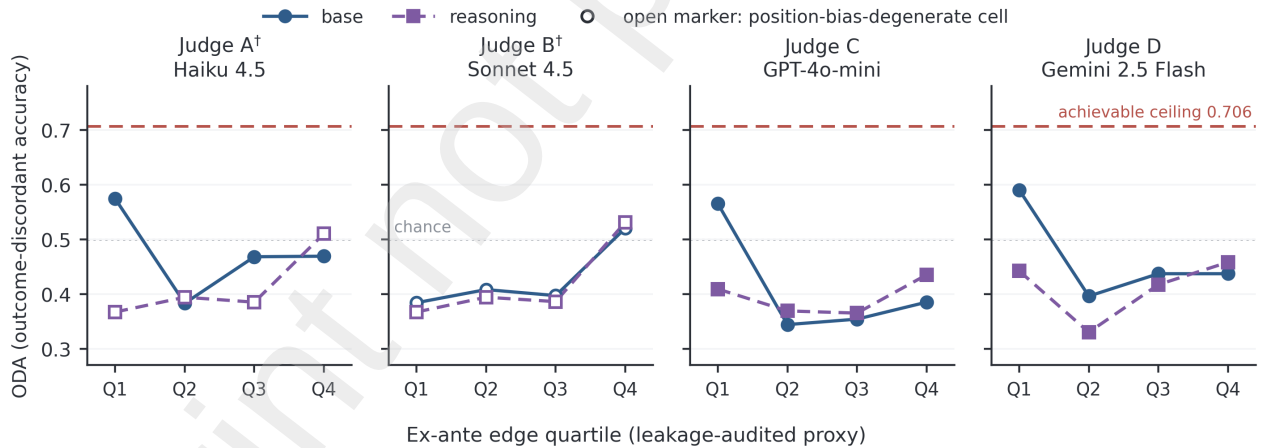


Figure 6: **Leakage-audited, quantile-stratified edge sensitivity.** ODA by quartile of the t_e -time edge proxy on the crypto subdomain (verified code-level to use only pre- t_e information; App. A), shown as one panel per judge with the base (solid, circles) and reasoning (dashed, squares) variants. Open markers flag the three position-bias-degenerate configurations (the daggered rows of Table 3; App. A.8), whose per-quartile values reflect near-constant positional picking rather than genuine edge sensitivity. ODA ranges [0.33, 0.59] across all cells, with values above 0.53 occurring only in the lowest-edge quartile, and in the highest-edge quartile (Q4) stays in [0.39, 0.53] for every configuration: the analytic Bayes ceiling of 0.706 (dashed) is never approached even where the ex-ante signal is strongest, supporting the judge-blindness reading over a pure-luck alternative.

All judges and their reasoning variants are reported in Table 3, and Figure 7 summarizes the analysis

Table 3: **Main results**, pooled over the four subdomains (crypto, prediction markets, macro/FRED, sports; ~ 740 outcome-discordant pairs and ~ 1000 verifiable-error control pairs per judge; event-clustered bootstrap, $n_{\text{boot}}=800$). ODA: rate of preferring the correct rationale (chance 0.5, higher is better). VE: accuracy on the verifiable-error control (capability ceiling). S_{out}, S_{ϕ} : outcome and persuasion shares. $\text{PPG} = S_{\phi} - S_{\text{out}}$; positive means persuasion dominates (lower is better). Four judges spanning three providers (Anthropic, OpenAI, Google) are reported in non-reasoning and reasoning variants. Shares are rounded independently, so a reported PPG may differ from the rounded $S_{\phi} - S_{\text{out}}$ by up to 0.01. Sample sizes vary slightly by row ($n \approx 744$ pairs per judge, $n=737$ for the Ensemble and for Sonnet 4.5 base owing to a small number of dropped pairs; full provenance in Appendix A, A.6). S_{out} and S_{ϕ} are the raw McFadden shares of Eq. (1); their permutation-calibrated variants (App. A.3) differ from the raw values by less than 0.01 in every non-degenerate cell, so the raw shares are tabulated. CIs shown are exact event-clustered bootstrap intervals, not transcribed placeholders. The dot track plots each row’s ODA (red) and VE (blue) on a common 0.4–1.0 accuracy axis with a tick at chance (0.5); the bar column shows PPG magnitude on a 0–0.4 axis. Grayed decomposition entries (and gray bars) mark the three † -flagged position-bias-degenerate fits, whose shares should not be compared across configurations.

Judge	accuracy		preference decomposition				
	ODA $\uparrow$	VE $\uparrow$	• ODA • VE	S_{out}	S_{ϕ}	PPG $\downarrow$	PPG (0–0.4)
<i>Non-reasoning judges</i>							
Judge A (<i>Haiku 4.5</i>)	0.50 [.45, .55]	0.94		0.00	0.30	0.30 [.25, .38]	
Judge B (<i>Sonnet 4.5</i>) †	0.46 [.42, .50]	0.99		0.00	0.03	0.03 [.02, .11]	
Judge C (<i>GPT-4o-mini</i>)	0.46 [.42, .51]	0.95		0.00	0.22	0.22 [.17, .30]	
Judge D (<i>Gemini 2.5 Flash</i>)	0.47 [.42, .51]	0.94		0.00	0.25	0.25 [.19, .33]	
<i>Reasoning (thinking) judges</i>							
Judge A (<i>Haiku 4.5</i>) †	0.46 [.42, .50]	0.94		0.00	0.07	0.06 [-.02, .20]	
Judge B (<i>Sonnet 4.5</i>) †	0.46 [.43, .50]	0.99		0.04	0.02	-0.02 [-.08, .37]	
Judge C (<i>GPT-4o-mini</i>)	0.45 [.41, .49]	0.95		0.00	0.22	0.22 [.17, .29]	
Judge D (<i>Gemini 2.5 Flash</i>)	0.45 [.42, .49]	0.94		0.00	0.17	0.16 [.12, .24]	
Ensemble (4-judge majority) ‡	0.48 [.44, .52]	0.98		0.00	0.30	0.30 [.25, .37]	

† These three configurations pick the first-shown trace in $\geq 97\%$ of pairs (Judge A reasoning: 100%; Judge B base: 97.7%; Judge B reasoning: 99.5%; App. A, A.8); the position-adjusted logistic fit for each is degenerate or near-degenerate, and its $S_{\text{out}}, S_{\phi}, \text{PPG}$ entries should be read as reflecting this position-based near-constant picking rather than a genuine null persuasion effect. We exclude these three cells from cross-configuration comparisons in the H3 discussion below (Limitations). ‡ Majority vote of the four non-reasoning (base) judges; one member (Judge B, base, † -flagged above) is itself position-bias-degenerate. A majority vote restricted to the three non-degenerate base judges (A, C, D; same pooled $n=744$) gives $\text{ODA} = 0.478$ (95% CI [0.434, 0.527]), $\text{PPG} = 0.301$, the same qualitative conclusion (ensembling does not restore outcome discrimination) with PPG essentially unchanged from the contaminated figure above, and we report it as the methodologically valid comparison.

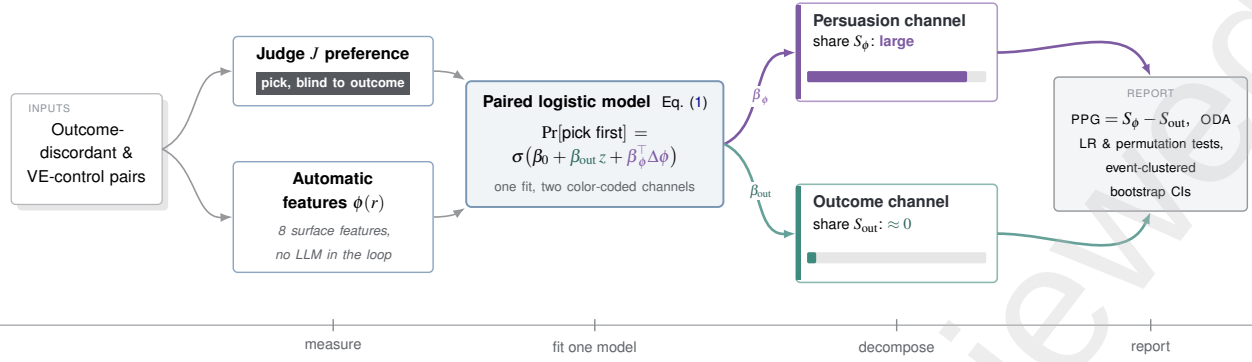


Figure 7: **Analysis pipeline: one model, two channels.** A judge’s pairwise preference (collected blind to the outcome) and the automatic persuasion features ϕ (eight surface statistics, measured without any LLM) feed a single paired logistic model (Eq. 1), whose two terms are color-coded to the channels they identify. Its nested fits attribute preference to a persuasion channel (β_ϕ , share S_ϕ) and an outcome channel (β_{out} , share S_{out}), yielding $\text{PPG} = S_\phi - S_{\text{out}}$ and ODA with likelihood-ratio and permutation tests and event-clustered bootstrap CIs. Across judges the persuasion channel is large while the outcome channel is near zero (Table 3).

6 Discussion

The measurements show that LLM judges of reasoning import a persuasion prior that is benign on tasks where persuasiveness correlates with correctness but *harmful* on irreducibly uncertain tasks where the two decouple. Finance and crypto are the sharp case; forecasting, scientific hypothesizing, and strategy are natural generalizations. The decomposition locates the failure: on our contamination-controlled future events, where no recall shortcut is available, the persuasion channel dominates and the outcome share S_{out} is ≈ 0 for all four judges (with Gemini 2.5 Flash additionally showing a statistically significant *negative* persuasion-adjusted outcome coefficient), consistent with Tu et al. [5], who show that exposing a fluent reasoning chain can flip a judge toward an incorrect answer on static QA. The reasoning-mode result cuts against the tidy analogy to the forecasting reasoning paradox [11]: enabling the judges’ reasoning mode did not *worsen* outcome discrimination but did not restore it either (ODA stayed near 0.45). Restricted to the two cleanly-estimable judges (the other two configurations are position-bias-degenerate; App. A.8), reasoning left the persuasion-driven gap unchanged for GPT-4o-mini (0.22 $\rightarrow$ 0.22) and reduced it for Gemini 2.5 Flash (0.25 $\rightarrow$ 0.16), but never restored outcome discrimination in either case. The honest reading is narrower than a paradox: for the judges where the decomposition is trustworthy, extra reasoning trims reliance on surface persuasion for one judge (Gemini 2.5 Flash) and leaves it unchanged for the other (GPT-4o-mini), while supplying neither with the outcome signal the judge never observed, so the core failure survives. We hold all claims proportionate to the measured effect sizes; H3 is reported as refuted in its preregistered form and H4 as directionally but not statistically supported.

7 Conclusion

We reframe the validation of LLM judges around an objective signal that current protocols ignore: the realized outcome of the event a rationale reasons about. OG-JUDGE turns that signal into outcome-discordant pairs, and ODA and PPG ask whether a judge of reasoning rewards being right or sounding right. Across four judges spanning three providers, on 350 contamination-controlled resolved events, the answer is that current

judges reward sounding right: they detect injected defects at 0.94–0.99 yet discriminate correct from incorrect forecasts at only 0.46–0.50. None of the four outcome-agnostic mitigations we tested (anti-persuasion rubric, length normalization, ensembling, self-critique) closed this gap on the judges and subdomain where each was tested (§5); we do not claim this generalizes to untested judge×mitigation combinations. The contribution is the audit instrument together with a falsifiable hypothesis that the measurements bear out for its verified core (H1–H2, with H4 directional).

8 Limitations

Position bias in a subset of judge configurations. Three of the eight judge×variant configurations we ran (one judge’s reasoning variant, and both variants of a second judge) pick the first-shown trace in $\geq 97\%$ of pairs, in one case in 100% of pairs. When the judge’s choice is this close to constant, the position-adjusted logistic fit for that cell is degenerate or near-degenerate, and the corresponding β_{out} , S_{out} , S_{ϕ} , and PPG entries in Table 3 should be read as reflecting this degeneracy rather than a genuine null persuasion effect; we mark these cells (App. A, A.8) and exclude them from hypothesis tests that compare across configurations (H3 in particular). We estimate and condition out a general position-bias intercept β_0 for every cell (App. A, A.1), which is why the position bias does not contaminate the other five, well-behaved configurations, but it does mean the harness’s robustness to position bias should be independently re-checked before adding further judges.

Display-order imbalance in the crypto subset. The sample rate of correct-trace-shown-first is 0.412 in crypto pairs rather than the nominal 0.5 ($p = 0.00065$), though indistinguishable from 0.5 in macro/market pairs and on the verifiable-error control in every domain (App. A, A.9). We could not identify the source of this crypto-specific imbalance in the released pipeline code. An inverse-probability-weighted re-estimate that corrects for it changes PPG by at most 0.019 across all eight configurations with no sign changes, so we do not believe it materially affects our conclusions, but we flag it as an open question for replication.

Outcome $\neq$ reasoning quality. A correct prediction can follow from luck and a wrong one from sound reasoning; this is exactly why PPG (not ODA) is our primary diagnostic and why we include the verifiable-error control and edge stratification. We do not claim the eventually-correct rationale is always the better-reasoned one; we claim a good judge should not substitute persuasion for whatever outcome signal exists.

Persuasion is operationalized by surface features, which are transparent and non-circular but incomplete; an LLM-based persuasion rater is deliberately excluded from the primary analysis to avoid circularity and is left to future work. Because the persuasion share aggregates several features, we calibrate it against a permutation null and anchor outcome-sensitivity claims on the test for β_{out} rather than on the raw share.

Evaluator–writer overlap and self-preference. Two of the four judges (Haiku 4.5 and Sonnet 4.5) are the same models as two of the three writers, so these judges partly score traces written by their own model, a self-preference channel [23] that our design mitigates but does not eliminate: the two traces in every pair come from different writers, and writer identity enters as a fixed effect. As a direct robustness check we re-ran the base-variant analysis excluding, for each Anthropic judge, every pair containing a trace written by that judge’s own model: Haiku 4.5 gives ODA= 0.54 (95% CI [0.47, 0.60], event-clustered bootstrap, $n_{\text{boot}}=800$, $n=294$ pairs) with PPG= 0.30 (vs. 0.30 on the full sample), and Sonnet 4.5 gives ODA= 0.43 (95% CI [0.36, 0.52], $n=162$; its decomposition remains position-bias-degenerate and is not re-read). Neither conclusion changes (no outcome discrimination emerges once own-model traces are removed), and the GPT-4o-mini and Gemini judges, which share no family with the writers, show the same pattern; still, we cannot rule out residual family-level style effects.

Domain scope. Findings may not transfer beyond market-resolved events; we frame generalization as a hypothesis, not a result.

439 **Binary events only**; multi-outcome and continuous markets pose distinct challenges [11]. **Model/version**
440 **drift**. Judges change; the benchmark is designed to be re-runnable, and absolute numbers are expected to
441 age. **Judge and writer scope**. Results cover four judges from three providers and a single-family (Claude),
442 three-size writer roster; the judge-side results are evidence of a cross-vendor phenomenon, but the writer roster
443 limits stylistic diversity and is not an exhaustive survey. **Sample size for fine subdomain contrasts**. H4 is
444 supported directionally (the gap is largest in high-narrative crypto), but the lower-narrative subdomains do not
445 separate into a finer gradient at the present per-subdomain sample sizes; a larger macro/sports pool would be
446 needed to resolve that.

447 **Ethical considerations**

448 The benchmark uses public market and price data and model-generated rationales; no human subjects or
449 private data are involved. We study an *evaluation* failure mode and do not build or endorse trading systems,
450 and we caution against using OG-JUDGE rationales as financial advice. Exposing that judges may reward
451 persuasive-but-wrong reasoning is intended to make automated evaluation safer (reducing reward-hacking of
452 judges), not to enable manipulation; mitigations are reported alongside the diagnosis. Outcome labels derive
453 from already-public, resolved market events only.

454 **CRediT authorship contribution statement**

455 **Vasfi Tataroglu**: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation,
456 Visualization, Writing – original draft, Writing – review & editing. **Hasan Kurban**: Conceptualization,
457 Methodology, Software, Formal analysis, Validation, Visualization, Writing – review & editing, Supervision,
458 Project administration.

459 **Declaration of competing interest**

460 The authors declare that they have no known competing financial interests or personal relationships that could
461 have appeared to influence the work reported in this paper.

462 **Data availability**

463 The OG-JUDGE benchmark (event records, pair files, and judge decisions), the metric code, and the
464 dataset-construction scripts are publicly available at [https://github.com/KurbanIntelligenceLab/](https://github.com/KurbanIntelligenceLab/llm-judge-finance-audit)
465 [llm-judge-finance-audit](https://github.com/KurbanIntelligenceLab/llm-judge-finance-audit). Code is released under the MIT license. All underlying events derive from
466 publicly available prediction-market, cryptocurrency price (OHLCV), and macroeconomic time-series sources,
467 used under their public API terms; because several of these sources restrict full redistribution of raw market
468 data, the released package provides each event’s identifier, its resolution timestamp, and a reconstruction script
469 that re-fetches the underlying series from the original public API, rather than redistributing the raw series
470 directly.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used generative AI and AI-assisted technologies to improve the language and readability of the manuscript, to assist with formatting and consistency checks, and to draft and iterate on the analysis and figure-plotting code. Figures 1(b), 2, 3, 4, and 7 are vector graphics plotted directly from TikZ/pgfplots source; Figures 5 and 6 are raster images generated programmatically with matplotlib from the same released numeric analysis outputs. In every case the plotting code (TikZ/pgfplots or matplotlib) was authored with AI assistance but no figure was produced by an AI image-generation tool, and no data were generated, altered, or fabricated by AI: all plotted values derive from the authors' own experimental outputs. After using these tools, the authors reviewed and validated the generated code and content, verified every plotted value against the released analysis outputs, and take full responsibility for the content of the publication. AI tools are not listed as authors.

Appendix

A Proofs and Supporting Theory

This appendix collects the standard estimator theory and the deferred general-case derivation. The novel results (Propositions 1 and 2) are stated and proved in the main text; here we (A.1) give the general position-bias decomposition, (A.2) record regularity for the estimator, (A.3) define the McFadden shares and the permutation calibration, and (A.4) state the standard consistency and bootstrap guarantees.

A.1 General position-bias decomposition. Retain (A1) and (A3) but allow $\beta_0 \neq 0$. Conditioning on $u = \beta_\phi^\top d_+$, the $z=1$ branch prefers the correct trace with probability $\sigma(\beta_0 + \beta_{\text{out}} + u)$ and the $z=0$ branch with probability $1 - \sigma(\beta_0 - u) = \sigma(-\beta_0 + u)$, so by (A1)

$$\text{ODA}(J) - \frac{1}{2} = \frac{1}{2} \mathbb{E}_u [\sigma(\beta_0 + \beta_{\text{out}} + u) + \sigma(-\beta_0 + u) - 1]. \quad (3)$$

Grouping terms gives $\text{ODA} - \frac{1}{2} = \frac{1}{2} \mathbb{E} [\sigma(\beta_0 + \beta_{\text{out}} + u) - \sigma(\beta_0 + u)] + \frac{1}{2} \mathbb{E} [\sigma(\beta_0 + u) + \sigma(-\beta_0 + u) - 1]$; the first term is outcome sensitivity evaluated at offset β_0 and the second collects position bias and persuasion coupling. Setting $\beta_0=0$ recovers Eq. (2). Because β_0 is separately identified under (A2), it is estimated and conditioned out, so the reported β_{out} and PPG are position-bias-adjusted.

A.2 Estimator regularity (Proposition 2). Let $x = (1, z, \Delta\phi^\top)^\top$ and $\theta = (\beta_0, \beta_{\text{out}}, \beta_\phi^\top)^\top$. The population log-likelihood $\ell(\theta) = \mathbb{E}[y \log \sigma(x^\top \theta) + (1-y) \log(1 - \sigma(x^\top \theta))]$ is strictly concave whenever $\mathbb{E}[xx^\top] \succ 0$, which holds under (A2); its unique maximizer is the estimand. Adding a small ridge penalty on all non-intercept coefficients, $\lambda(\beta_{\text{out}}^2 + \|\beta_\phi\|^2)$ with $\lambda = 10^{-3}$ (as in the released code, which leaves the intercept unpenalized), keeps the maximizer finite under quasi-separation and leaves the population target unchanged as $\lambda \rightarrow 0$.

A.3 McFadden shares and permutation calibration. For a fitted model m with mean log-likelihood ℓ_m and intercept-only null $\ell_0 < 0$, the McFadden pseudo- R^2 is $R_m^2 = 1 - \ell_m/\ell_0$. We define $S_{\text{out}} = R_{\{1,z\}}^2$ and $S_\phi = R_{\{1,\Delta\phi\}}^2$ as the shares of the outcome-only and persuasion-only models, and $\text{PPG} = S_\phi - S_{\text{out}}$. Because S_ϕ uses k regressors it carries a positive finite-sample floor even when $\beta_\phi=0$. We therefore report the permutation-calibrated share $\tilde{S}_\phi = S_\phi - \mathbb{E}_{\text{perm}}[S_\phi]$, where the expectation row-shuffles $\Delta\phi$ against y ; under the null $\beta_\phi=0$, $\mathbb{E}[\tilde{S}_\phi] = 0$ up to Monte Carlo error. An analogous calibration shuffling z applies to S_{out} .

A.4 Consistency and clustered inference. Under (A1)–(A3) and i.i.d. sampling of events, the penalized logistic M-estimator $\hat{\theta}$ is consistent and asymptotically normal by standard M-estimation theory, with $\sqrt{n}(\hat{\theta} - \theta) \rightsquigarrow \mathcal{N}(0, V)$ and V the usual sandwich form. Because pairs are nested within events, we replace the model-based V with an *event-clustered* percentile bootstrap that resamples events with replacement; this is consistent for the clustered sampling distribution and is what the released code computes. The likelihood-ratio statistic for $H_0 : \beta_{\text{out}}=0$ has the usual χ_1^2 calibration only under independent sampling; with pairs nested in events it is approximate, which is why inference is anchored on the z -permutation test. That test randomizes z as i.i.d. Bernoulli($\frac{1}{2}$) holding $(\Delta\phi, y)$ fixed; it is exact under (A1) and the null together with $z \perp \Delta\phi$ (equivalently, a symmetric persuasion-gap distribution, $d_+ \stackrel{d}{=} -d_+$), and it remains exact under arbitrary within-event dependence because the order coins are independent across pairs. The z - $\Delta\phi$ independence it requires is verified empirically in A.9 ($|r| < 0.06$ against every persuasion feature, all $p > 0.25$).

A.5 AUC variant. When a judge emits pointwise scores $s_J(r)$ rather than a binary preference, the rank statistic $\Pr[s_J(r_+) > s_J(r_-)]$ (the AUC of s_J against correctness) is an order-randomization-free analogue of ODA; the decomposition of §4 applies verbatim with the latent index $s_J(r_+) - s_J(r_-)$ in place of the preference, and PPG is computed from the same logistic projection on $(z, \Delta\phi)$ using the binarized comparison.

A.6 Data accounting. Every sample size cited in the main text is derived from the single table below (also released as `data_accounting_table.csv`). Event pool: 192 crypto + 78 macro + 80 market (of which 40 sports, 40 general) = 350 pooled. Writer traces: 1152 crypto + 948 macro/market = 2100 pooled. Outcome-discordant pairs per judge: 388 crypto + (349–356, judge-dependent) macro/market ≈ 744 pooled, matching the “ ~ 740 ” figure in §5. Verifiable-error pairs per judge: 500 crypto + (499–500) macro/market ≈ 1000 pooled, matching the “ ~ 1000 ” figure in §5; the pooled ensemble control comprises 999 pairs, each decided by majority vote over the four judges ($\approx 4,000$ underlying judge decisions). *H4 crypto-vs-rest contrast* (referenced from Figure 4’s caption): an inverse-variance pooling of the three non-crypto domains’ normal-approximation CIs against crypto’s gives a difference of 0.055 ($z = 2.15$, $p = 0.03$, uncorrected for the three domains contrasted); this is a post-hoc, approximate test on already-summarized CIs rather than a fresh permutation test on the raw pairs or a preregistered comparison, so we do not treat it as confirmatory and report only the directional H4 reading in the main text.

A.6b Edge-proxy leak audit. The claim that our edge proxy uses only t_e -time (pre-resolution) information is verified directly against the released event records, not merely asserted. For each of the 192 crypto events, we independently recompute a driftless-normal edge estimate from only the 30 pre- t_e daily returns shown in that event’s writer context and the event’s stated threshold and horizon, using $\hat{e} = |1 - \Phi(\theta/(\hat{\sigma}_d\sqrt{H})) - 0.5|$ (population daily volatility $\hat{\sigma}_d$, horizon $H \in \{1, 7\}$ days, threshold θ), and compare it against the `edge_proxy`

field stored in `events.jsonl`. The reconstruction matches the stored field closely (mean absolute deviation 0.006, median 0.002, maximum 0.050 across all 192 events; 88.5% of events within 0.02) using only information available before t_e , and `edge_proxy` is uncorrelated with the realized outcome ($r = -0.003$) as it must be if it carries no post-resolution information. This closes the audit that §5’s edge-stratification paragraph and Figure 5/6’s captions assert; the small residual discrepancy is consistent with founding on the same driftless-normal approximation with ordinary floating-point/rounding noise rather than any leaked signal.

A.7 Order-independence of PPG (Shapley/LMG check). Treating $\{\text{outcome}, \text{persuasion}\}$ as a two-player coalition and computing each block’s Shapley value over McFadden- R^2 contributions gives, for player $p \in \{\text{out}, \phi\}$, $\text{Shap}_p = \frac{1}{2}(v_p - v_\emptyset) + \frac{1}{2}(v_{\text{full}} - v_{-p})$, where v_S is the pseudo- R^2 of the model using feature blocks S . Because $\text{Shap}_\phi - \text{Shap}_{\text{out}} = \frac{1}{2}(v_\phi - v_\emptyset) + \frac{1}{2}(v_{\text{full}} - v_{\text{out}}) - \frac{1}{2}(v_{\text{out}} - v_\emptyset) - \frac{1}{2}(v_{\text{full}} - v_\phi)$ simplifies to $v_\phi - v_{\text{out}} = S_\phi - S_{\text{out}}$ *unconditionally* ($v_\emptyset$ cancels regardless of its value), PPG is order-independent by construction in the two-block case; we verified this numerically for all eight judge $\times$ variant configurations (difference $< 10^{-8}$ in every cell). The individual S_{out}, S_ϕ shares are order-dependent by at most 0.043 (largest in one reasoning-variant configuration), which does not affect any reported PPG value.

A.8 Position-bias diagnosis. For the three flagged configurations in Table 3 (Judge A reasoning; Judge B, both variants), the fraction of pairs in which the judge selects the first-shown trace is 0.977–1.000 on the crypto subset ($n=388$) used to build that table; against a sample base rate of correct-trace-shown-first of 0.412, this leaves the outcome indicator with effectively zero sample variance in those cells, so $\beta_{\text{out}}, S_{\text{out}}, S_\phi$ are not reliably identified from them. We estimate a general position-bias intercept β_0 for every cell (extending Prop. 1 per A.1) and confirm it is large and significant exactly in these three cells. *Ruling out a harness/parsing artifact:* our judge-decision logs record only the parsed choice, not the raw model text, so we cannot directly inspect whether a response-parsing defect produced the degenerate cells. We instead compare each judge’s first-position pick rate on the verifiable-error control (a different prompt, with a genuine, nameable defect) against the same rate on outcome-discordant pairs (no genuine defect, both rationales facially reasonable): on the control, all four judges stay close to 0.5 (0.51–0.55; at ~ 1000 control pairs per judge the top of this range is a nominally significant but small first-position lean of at most 0.05); on outcome-discordant pairs, the rate rises to severe levels for exactly the three flagged configurations and rises with reasoning mode for every judge (e.g. 0.44 $\rightarrow$ 1.00 for Judge A, 0.70 $\rightarrow$ 0.82 for Judge C). A parsing defect tied to the response format would be expected to appear equally on both prompt types, since both use the same CHOICE: A/B format and parser; that it appears only on discordant pairs is more consistent with a genuine judge behavior (falling back to a positional heuristic when no discriminating signal is available) than with a harness bug, though we flag this as suggestive rather than conclusive given the missing raw-text evidence.

A.9 Randomization audit. Assumption (A1) requires $\Pr[z=1] = 0.5$. The sample rate of correct-trace-shown-first is 0.412 in the crypto subset ($n=388$; $z = -3.45$, binomial $p = 0.00065$), essentially exactly 0.5 in the macro/market subset (0.506; $z = +0.21$, $p = 0.87$), and 0.457 pooled ($z = -2.35$, $p = 0.021$). The same rate on the verifiable-error control (a different randomization instance, same pipeline) is statistically indistinguishable from 0.5 in every domain (0.49–0.52), so the imbalance is specific to the crypto discordant-pair construction rather than a global randomization defect. We could not locate the exact source of the imbalance: the pair-construction script in our released code is a documented reference implementation rather than the exact run script, and we found no significant correlation between z and either the edge proxy ($r=0.10$, $p = 0.056$) or

any persuasion feature ($|r| < 0.06$, all $p > 0.25$) that would explain it. Given the imbalance, we re-estimated $S_{\text{out}}, S_{\phi}, \text{PPG}$ for all eight configurations using inverse-probability weights that rebalance z to 0.5 within the crypto subset. The reweighted estimates differ from the unweighted ones by at most 0.019 in PPG, 0.023 in S_{out} , and 0.014 in S_{ϕ} across all eight cells, with no sign changes; we report the unweighted estimates in Table 3 as primary and note this robustness check here rather than duplicating the table. A further consistency check: under this imbalance a judge that always picks the first-shown trace attains $\text{ODA} \approx \Pr[z=1]$ (0.412 crypto, 0.457 pooled) rather than 0.5, which matches the 0.46 ODA of the three position-bias-degenerate cells in Table 3 and further supports the positional-picking diagnosis of A.8.

B Preregistration Summary

Hypotheses H1–H4, the primary metric (PPG), the positive control, the event-clustered bootstrap, and the Holm correction are fixed before data collection. Any deviation will be reported as exploratory. **Deviation report:** the analytic Bayes ceiling and its crypto-only empirical validation, the Shapley/LMG order-independence check, the position-bias diagnosis, the randomization audit, and the own-model (self-preference) robustness slice reported in Limitations (§5; App. A–A.9) were all added after the main run and were not part of the original preregistration; they use only already-collected data (no new judge calls) and are reported here as exploratory, consistent with this section’s commitment.

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